

Professional Responsibilities of Engineers

Ethics and Accreditation

- An accredited degree course must include some discussion of ethics:
- “[students] must appreciate the social, environmental, ethical, economic and commercial considerations affecting the exercise of their engineering judgement” (UK-SPEC)

Learning about ethics helps students to:

- Develop critical thinking skills and *judgement*
- *Understand* practical difficulties and use suitable approaches and techniques to help people produce better outcomes
- *Develop* an ethical identity to carry forward to their working life

Learning about ethics helps students to:

- *Understand* the nature of professional and personal responsibility
- Be able to *identify* the ethical element of any decision
- Be able to *address* problems arising from questionable practice using appropriate tools

RAE and EPC

Teaching Engineering Ethics Group

- Curriculum Map
 - Aimed at *undergraduate* teaching
 - A *guide*, not a prescription
 - Provides a *framework*, not rules

Sheffield principle

- Based on the notion that ethical problem solving for engineers is much like any design problem:
 - Multiple workable solutions
 - Needs a framework for finding a suitable solution

Theme

- What dilemmas are you likely to face as a young engineer?
 - What is it to be a professional?
 - To whistleblow or not?
 - Health and Safety in practice
 - Whether to accepting a gift

Topics

- Expectations of an engineer:
 - Is there a social contract between professionals and society?
 - What are your obligations to society?
 - What do you expect in return?

Topics

- Individual responsibilities:
 - How do you deal reliably and robustly with ethical dilemmas?
 - Philosophical principles – Consequentialism, Duty Ethics, Casuistry
 - Ethical decision making – where to draw the line

Topics

- Dealing with disasters:
 - Health and Safety responsibilities
 - Risks and hazards
 - Whistleblowing

Example: Being a Professional

- What are society's expectations of
 - a medical doctor?
 - a professional engineer?
- Discuss in groups
- Critical essay from each group
- Peer evaluation of contribution to group work

Gifts and Bribes example

Denise is an engineer at a large manufacturing company. It is her job to specify gears for a new product. After some research and testing, she decides to use ACME gears for the job.

The day after she places a £100,000 order for gears, an ACME representative visits her and gives her a voucher for an all-expenses-paid trip to the ACME Technical Forum in Jamaica.

The 4-day trip will include 18 classroom hours and a day-long tour of the coastline.

Does this trip fall under the category of bribery?

What should Denise do?

Method of Casuistry

1. State problem as an “or” question.
 - Is this trip to Jamaica a bribe or not?
 - Is this trip a case of bribery or a morally unproblematic gift?
2. Set up paradigms
 - a. Set up ends of spectrum:
 - Positive Paradigm: case that is clearly ok
 - Negative Paradigm: case that is clearly not ok
 - look at “or” question
 - b. List features of Positive Paradigm ... costs £1
 - c. List corresponding features of Negative Paradigm...costs £5,000

Method of Casuistry

3. Mark which features are the most important
4. Compare paradigms with Test Case on each of the features
5. Judge which paradigm Test Case is closest to
6. State Conclusion - Trip is/is not a case of bribery

Is the trip to Jamaica a gift or a bribe?

Feature	Positive Paradigm	Test Case	Negative Paradigm
Creates an obligation	No		Yes
Denise's decision making power	None		Sole
Timing	After decision		Before decision
Reason	Education		Pleasure
Quality of Product	Best		Worst
Gift size	£1.00		£5,000

Different to other engineering modules

- Scope for:
 - Inter-cultural awareness in group work
 - Dealing with uncertainty and ‘grey’ areas
 - Recognising that there is rarely a ‘right’ answer